

1 Multi-Armed Bandit with Decreasing and Constant Step Sizes

For the multi-arms bandit problem we discussed in the class. Suppose that we get return G_n at n -th time we do action a , and $\mathbb{E}G_n = r$, $n = 1, 2, \dots$.

Let Q_{n+1} be our estimates of r after we do action a the n -th time, and we have the following update rule

$$Q_{n+1} = Q_n + \alpha_n(G_n - Q_n), \quad Q_1 = 0.$$

We define

$$V_n = \mathbb{E}[(Q_n - r)^2].$$

(a) (Decreasing step size) Let $\alpha_n = \frac{1}{n}$, show that

(i) (5 points)

$$Q_{n+1} = \frac{1}{n} \sum_{i=1}^n G_i, \quad n = 1, 2, \dots$$

$$\alpha_n = \frac{1}{n}, \therefore Q_{n+1} = Q_n - \frac{1}{n}Q_n + \frac{1}{n}G_n$$

$$\text{So } nQ_{n+1} - (n-1)Q_n = G_n$$

$$\text{Let } A_{n+1} = nQ_{n+1}, \quad A_1 = 0, \text{ Then } A_{n+1} - A_n = G_n$$

Cumulative sum from 1 to n ,

$$A_{n+1} - A_1 = \sum_{i=1}^n G_i$$

$$\therefore Q_{n+1} = \frac{1}{n} \sum_{i=1}^n G_i$$

(ii) (10 points)

$$\lim_{n \rightarrow \infty} V_n = 0.$$

$$V_n = \mathbb{E}[(Q_n - r)^2] = \mathbb{E}\left[\left(\frac{1}{n} \sum_{i=1}^n G_i - r\right)^2\right]$$

G_i means the return we get after i -th we do action a . So when $n \rightarrow \infty$, $\frac{1}{n} \sum_{i=1}^n G_i \rightarrow$ the expected value of $G_i = E[G_i] = r$

$$\therefore \lim_{n \rightarrow \infty} V_n = E[(r - r)^2] = 0$$

(b) (Constant step size) Let $\alpha_n = \alpha$, $0 < \alpha < 2$, show that

(i) (15 points)

$$V_{n+1} = (1 - \alpha)^2 V_n + \alpha^2 \text{Var}[G_n],$$

where

$$\text{Var}[G_n] = \mathbb{E}[(G_n - r)^2].$$

$$Q_{n+1} = Q_n + \alpha(G_n - Q_n)$$

$$\text{So } Q_{n+1} = (1 - \alpha)Q_n + \alpha G_n \quad \textcircled{1}$$

$$\text{We have } E(G_n) = r, \text{Var}(G_n) = E[(G_n - r)^2] \text{ and } V_n = \mathbb{E}[(Q_n - r)^2]$$

From $\textcircled{1}$,

$$Q_{n+1} - r = (1 - \alpha)Q_n + \alpha G_n - r = (1 - \alpha)(Q_n - r) + \alpha(G_n - r)$$

$$\therefore E[(Q_{n+1} - r)^2] = E[((1 - \alpha)(Q_n - r) + \alpha(G_n - r))^2]$$

$$= E[(1 - \alpha)^2(Q_n - r)^2] + E[\alpha^2(G_n - r)^2] + E[2(1 - \alpha)\alpha(Q_n - r)(G_n - r)]$$

$$= (1 - \alpha)^2 V_n + \alpha^2 \text{Var}(G_n) + 2(1 - \alpha)\alpha E[Q_n - r] \cdot 0$$

$$\text{So } V_{n+1} = (1 - \alpha)^2 V_n + \alpha^2 \text{Var}(G_n)$$

(ii) (20 points)

$$\lim_{n \rightarrow \infty} \left| V_{n+1} - \frac{\alpha}{2 - \alpha} \text{Var}[G_n] \right| = 0.$$

Because V_n is the value function. So $n \rightarrow \infty$, $V_{n+1} \rightarrow V$

$$\therefore V = (1 - \alpha)^2 V + \alpha^2 \text{Var}[G_n]$$

$$\text{We get: } V = \frac{\alpha^2 \text{Var}[G_n]}{1 - 1 - \alpha^2 + 2\alpha} = \frac{\alpha}{2 - \alpha} \text{Var}[G_n]$$

$$\lim_{n \rightarrow \infty} V_n = \frac{\alpha}{2 - \alpha} \text{Var}[G_n]$$

$$\text{So } \lim_{n \rightarrow \infty} \left| V_{n+1} - \frac{\alpha}{2 - \alpha} \text{Var}[G_n] \right| = V - \frac{\alpha}{2 - \alpha} \text{Var}[G_n] = 0$$

2 ϵ -greedy Algorithm and Long-Run Performance

Recall that the ϵ -greedy algorithm do the following:

$$A_t = \begin{cases} \arg \max_a Q_t(a) & \text{with probability } 1 - \epsilon, \\ \text{choose } a \text{ uniformly at random} & \text{with probability } \epsilon, \end{cases}$$

where

$$Q_t(a) = \frac{1}{N_t(a)} \sum_{k=1}^t R_k \mathbf{1}_{A_k=a}, \quad N_t(a) = \sum_{k=1}^t \mathbf{1}_{A_k=a}.$$

Let

$$\mathbb{E}[R_t \mid A_t = a] = q_a, \quad a^* = \arg \max_a q_a.$$

- (a) (5 points) Explain briefly why $\epsilon = 0$ is worse than $\epsilon > 0$ in the long run.

Because $\arg \max_a Q_t(a)$ selects the action with the highest current estimated value with greedy algorithm, it may not be the true optimal action in the long run performance. when $\epsilon = 0$, it will stop explore and keep exploiting that wrong action forever.

- (b) (10 points) Show that

$$\mathbb{E}R_t = \sum_a q_a P(A_t = a).$$

$$\because E[R_t \mid A_t = a] = q_a$$

So in order to calculate the expected value of R_t , we need to include all Action.

$$\therefore E[R_t] = \sum_a E[R_t \mid A_t = a] \cdot P(A_t = a) = \sum_a q_a P(A_t = a)$$

- (c) (15 points) With $\epsilon > 0$, suppose at time T we have

$$\arg \max_a Q_T(a) = a^*,$$

compute $\mathbb{E}R_T$. How does $\mathbb{E}R_T$ change as we increase ϵ , the exploration rate?

$\because \epsilon > 0$, $\therefore$ we choose a^* with probability $1 - \epsilon$, and choose action randomly with probability ϵ .

Then $E[R_t] = q_{a^*}(1 - \epsilon) + E[R_t | A_t]\epsilon$

$$= q_{a^*}(1 - \epsilon) + \frac{1}{K} \sum_a q_a \epsilon$$

(K is the total number of actions)

As ϵ increases, $E[R_t]$ decreases, because more probability is assigned to random exploration instead of selecting the best action a^* .

(d) (20 points) Answer Sutton & Barto Exercise 2.3:

In the comparison shown in Figure 2.2, which method will perform best in the long run in terms of cumulative reward and probability of selecting the best action? How much better will it be? Express your answer quantitatively.

$\epsilon = 0.1$ performs best in the long run in both average reward and probability of optimal action.

The average reward increases about 0.4 compared to greedy algorithm and about 0.1 compared to $\epsilon = 0.01$.

And the % optimal action increases about 45% compared to greedy algorithm and about 30% compared to $\epsilon = 0.01$.

It shows moderate exploration will not avoid losses in long run.

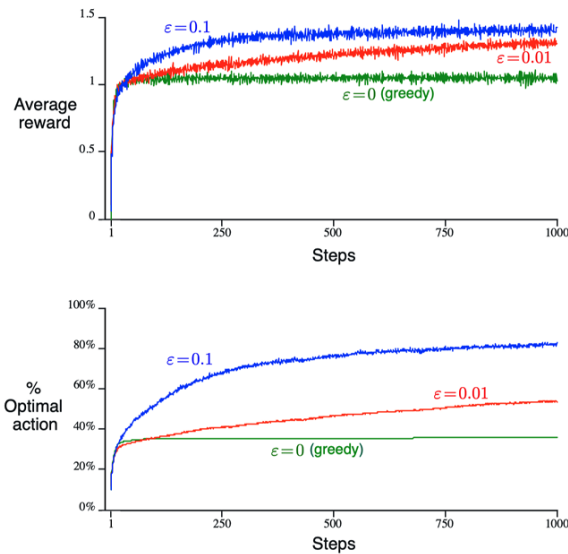


Figure 2.2: Average performance of ϵ -greedy action-value methods on the 10-armed testbed. These data are averages over 2000 runs with different bandit problems. All methods used sample averages as their action-value estimates.